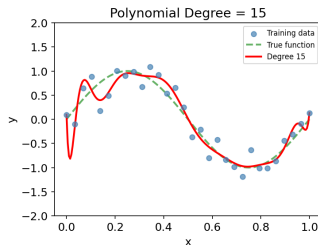
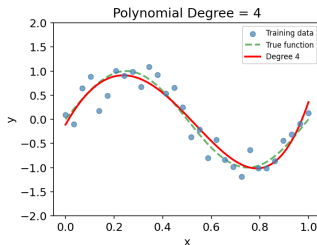
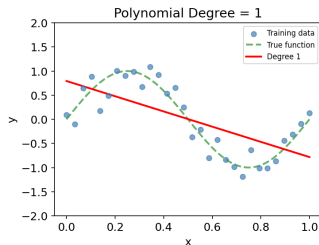


Overfitting and Underfitting in Machine Learning

AIAA 2711 — Spring 2025

Member A Member B Member C

The Hook: Which Model Do You Trust?



Which polynomial would you trust to predict a new data point?

Vote by show of hands: degree 1, degree 4, or degree 15?

Formal Definitions

Training Error:

$$E_{\text{train}} = \frac{1}{N} \sum_{i=1}^N \ell(y_i, \hat{f}(x_i))$$

Test Error:

$$E_{\text{test}} = \mathbb{E}_{(x,y) \sim P} [\ell(y, \hat{f}(x))]$$

Underfitting E_{train} high — model too simple

Overfitting $E_{\text{train}} \ll E_{\text{test}}$ — generalization gap

Key Message

The **generalization gap** is our diagnostic.

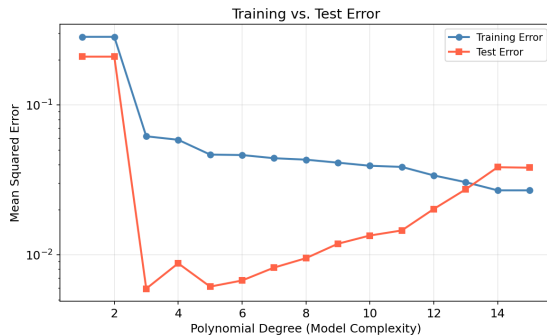
The Key Question: What Shapes the Error Curve?

- Training error $\downarrow$ monotonically with complexity
- Test error is **U-shaped**
- Minimum near degree 3–5

The answer:

Bias-Variance Decomposition

(uses expectation & variance from Weeks 9–10)



Bias-Variance Setup

Data model:

$$y = f(x) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

- $f(x)$: true underlying function
- $\hat{f}(x)$: model trained on dataset D

Goal: decompose

$$\mathbb{E}_D[(y - \hat{f}(x))^2]$$

Key Point

The expectation is over all possible training datasets D .

Bias-Variance Derivation

Step 1: Add-subtract $f(x)$:

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$$\mathbb{E}_D[(y - \hat{f})^2] = \mathbb{E}_D[(f - \hat{f})^2] + \sigma^2$$

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Step 2: Add-subtract $\mathbb{E}_D[\hat{f}]$:

$$\mathbb{E}_D[(f - \hat{f})^2] = (f - \mathbb{E}_D[\hat{f}])^2 + \mathbb{E}_D[(\hat{f} - \mathbb{E}_D[\hat{f}])^2]$$

Cross-term vanishes: $\mathbb{E}_D[\hat{f} - \mathbb{E}_D[\hat{f}]] = 0$

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Final Result

$$\text{Error} = \underbrace{(f - \mathbb{E}_D[\hat{f}])^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}_D[(\hat{f} - \mathbb{E}_D[\hat{f}])^2]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible}}$$

Verified via Monte Carlo simulation — see figure in backup.

Regularization: Constraining Complexity

High variance $\rightarrow$ overfitting. **Fix: constrain model complexity.**

OLS (no penalty):

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$$

Ridge (ℓ_2 penalty):

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_2^2$$

Note

The ℓ_2 norm is the Euclidean norm from Week 3.

Ridge: Closed-Form & Interpretations

Closed-form solution:

$$\mathbf{w}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

vs. OLS: $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$

Two interpretations:

- ① **Eigenvalue (Week 5):** $\alpha \mathbf{I}$ shifts eigenvalues $\lambda_i \rightarrow \lambda_i + \alpha$, prevents instability when $\lambda_i \approx 0$.
- ② **Lagrangian (Week 9):** Equivalent to $\min \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2$ s.t. $\|\mathbf{w}\|^2 \leq t$; α is the Lagrange multiplier.

Key Insight

Adding $\alpha \mathbf{I}$ trades a small bias increase for a large variance decrease.

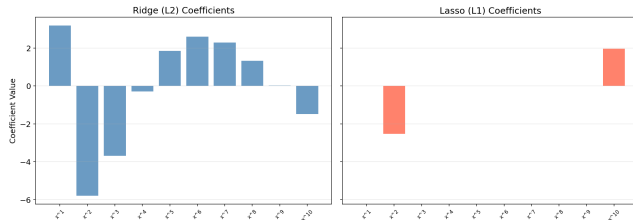
Lasso & Geometric Comparison

Lasso (ℓ_1 penalty):

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1$$

Geometric intuition:

- ℓ_2 ball (circle) $\rightarrow$ Ridge $\rightarrow$ smooth shrinkage
- ℓ_1 ball (diamond) $\rightarrow$ corners on axes $\rightarrow$ **sparse solutions** (exact zeros)



Key insight: Lasso = regularization + feature selection.

Cross-Validation: Choosing α

How to choose α ? K -fold cross-validation.

- 1 Partition data into K folds
- 2 Train on $K-1$ folds, evaluate on fold k
- 3 Average K scores
- 4 Select α minimizing CV error

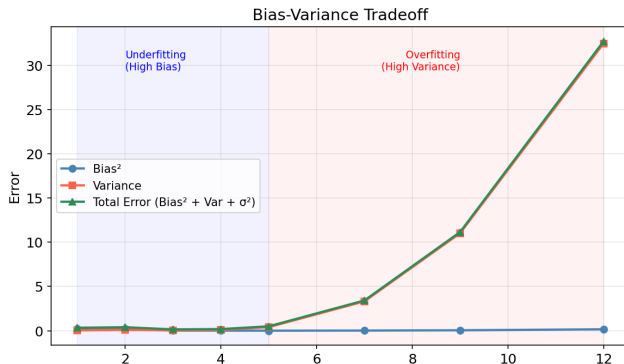
Key Point

Unbiased test error estimate without a separate validation set.

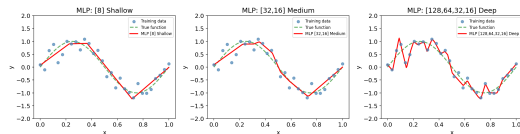
Experimental Verification

Theory → Experiment mapping:

Theory Prediction	Method	Result
U-shaped test error	Polynomial fits	Confirmed
Bias↓ Variance↑ with complexity	Monte Carlo (200 datasets)	Confirmed
Ridge shrinks coefficients	Ridge regression	Confirmed



Deep Learning & Summary



Same bias-variance principle applies to neural networks.

3 Key Takeaways:

- 1 **Bias-Variance decomposition** explains the tradeoff mathematically
- 2 **Regularization** (Ridge ℓ_2 / Lasso ℓ_1) controls complexity via norm penalties
- 3 **Cross-validation** selects the optimal balance point

Scenario: A model achieves 99% training accuracy but only 60% test accuracy.

- **Q1:** Overfitting or underfitting?
- **Q2:** What is your fix?

Interactive Closing & Q&A

Scenario: A model achieves 99% training accuracy but only 60% test accuracy.

- **Q1:** Overfitting or underfitting?
- **Q2:** What is your fix?

Answer: Generalization gap = 39% → **Overfitting**
→ Add regularization / reduce model complexity

Thank You — Questions?